# DIRECTRF

USER GUIDE | Nearby phone-to-phone calling

DirectRF uses Google Nearby Connections to find and connect two nearby Android phones over Bluetooth/Wi-Fi radios. No app signaling server, account, or internet connection is part of the call path. Google Play services is required on both phones.

**Real-device status: UNVERIFIED. This build has not been run on two physical phones. Discovery, pairing, two-way audio, screen-off behavior, reconnection, range, and latency are not yet confirmed.**

## Requirements

- Two Android phones running Android 10 (API 29) or later, each with Google Play services.
- Bluetooth and Wi-Fi radios enabled; working microphone and speaker/headset.
- Nearby permissions for discovery and microphone permission when calling.
- The same APK installed on both phones.

## Connect two phones

1. Open DirectRF on both phones and enter a different local display name on each.
2. Open Devices on both and tap Find nearby phones. Grant the requested Nearby permissions.
3. On one phone, tap Connect on the other phone’s name.
4. Compare the authentication digits on both screens. Accept only if they match.
5. After both screens show connected, place a call. Accept or decline on the receiving phone.

## During a call

Use Mute, Speaker/Earpiece, and End call. The foreground notification indicates an active call. If Android reports microphone or Opus setup failure, the call should show an error instead of entering active state.

## Range and limitations

Nearby phone-to-phone range is short. Roughly 10-100 m is an estimate under suitable conditions, not a measurement or guarantee from this project. Walls, interference, handset models and the environment can reduce it. This uses ordinary phone Bluetooth/Wi-Fi, not custom RF hardware or the cellular modem.

**Follow docs/MANUAL_TWO_PHONE_TEST.md for exact pass/fail checks and record device-specific results before relying on the app. The script is currently marked NOT RUN.**